

LORAD: Robust Optical Relay Decoding for Constrained Lunar Surface Proximity Links

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Abstract—Short-range infrared links are attractive where radio-frequency operation is constrained, but simple embedded receivers typically decode them with hand-tuned amplitude or timing thresholds that discard a symbol, and with it an entire byte, whenever a pulse falls outside its acceptance window. This paper presents LORAD, an experimental optical proximity link, and shows that replacing threshold decoding with maximum-likelihood sequence detection substantially improves link reliability without any change to the transmitter, the receiver electronics, or the modulation.

The link uses baseband on-off keying in which symbol identity is carried by pulse width rather than amplitude, a choice validated by measurement: with the receiver's transimpedance stage saturated, peak amplitude held to within 0.17% across every pulse recorded while pulse-width classes remained separated by 12–40 pooled standard deviations. A dual-photodiode receiver streams per-pulse observations to a decoder that combines a learned emission model, inter-pulse gap timing, and the protocol's own byte-framing and printable-ASCII constraints in a hidden Markov model decoded by the Viterbi algorithm.

The system is characterized over five data rates from 28.6 to 171.4 bit/s using 597 labeled transmissions, with a further 157 transmissions decoded in a live closed loop on hardware. Held out by message, the sequence decoder raises the transmission success rate from 72.2% to 100% at 28.6 bit/s and from 36.1% to 69.4% at 114.3 bit/s. In live operation at 114.3 bit/s it raises transmission success from 5.9% to 44.9% and halves character error rate, recovering 41% of the transmissions the threshold decoder lost. Analysis shows the dominant failure mode is undetected pulses rather than misclassified ones, which per-symbol classifiers cannot address by construction, and that every such loss leaves a recoverable signature in the inter-pulse gap.

Index Terms—Optical wireless communication, proximity links, maximum-likelihood sequence detection, hidden Markov models, embedded receivers, lunar surface operations

I. INTRODUCTION

Surface operations on the Moon will depend on short-range links between assets that are close together but separately powered: a rover and its lander, a sensor mast and a relay node, an instrument package and the vehicle that deployed it. These *proximity* links carry housekeeping telemetry, commands, and instrument data over metres rather than kilometres, and they

are not the same design problem as the high-rate optical trunk links that dominate the space optical communication literature.

Optical signalling is attractive at this scale for reasons specific to the environment. It requires no spectrum allocation and generates no radio-frequency emissions that might couple into sensitive instrumentation. It is inherently directional, which limits mutual interference between co-located assets. Emitter and photodiode front ends are light and draw little power, which matters where every gram and every watt is contested. Permanently shadowed regions near the lunar poles, of particular interest for volatiles prospecting, additionally offer very low ambient infrared background, an operating condition that favours optical reception and is difficult to reproduce terrestrially.

The constraint that shapes such a link is not aperture or transmit power but the receiver. A surface asset's communication processor is small, radiation-tolerant, and shared with other tasks, so decoding must be cheap enough to run inside a sampling loop. In practice this means threshold decoding: each detected pulse is assigned to a symbol by testing it against fixed acceptance windows. That approach has a failure mode which is rarely quantified. It has no representation for a pulse that was never detected at all, so when the receiver misses a symbol the byte containing it is lost, and because protocols of this kind spend several pulses per byte, a small per-symbol error rate compounds into a large message error rate. Improving the per-symbol classifier cannot help, because the missing observation was never presented to it.

Message loss is more costly in this setting than the raw numbers suggest. Proximity links operate on constrained duty cycles and within scheduled communication windows, so a lost message is not simply retransmitted at negligible cost: retransmission consumes energy, occupies window time, and on a store-and-forward relay path can delay delivery by an entire orbit. Recovering a message at the receiver, using information already present in the captured signal, is therefore worth more than the same improvement would be on a terrestrial link with cheap retries.

This paper presents LORAD (Lunar Optical Relay And Decoding), an experimental platform built to measure that effect and to address it. The link uses baseband on-off keying in which symbol identity is carried by pulse *width* rather

This paper was completed as part of classwork for Application of Sensing Systems at Johns Hopkins Whiting School of Engineering taught by Elishiah Miller, Ph.D. and was funded by NASA's Advanced Degree Program.

than amplitude. That choice is not incidental. Measurements presented in Section V show the receiver’s transimpedance stage to be saturated, pinning received peak amplitude to within 0.17% across every pulse recorded, so an amplitude-discriminating scheme could not function on this hardware, whereas pulse-width classes remain separated by 12–40 pooled standard deviations even at rates where decoding fails outright. Time-domain encoding is also the more defensible choice for a dusty, thermally cycling surface environment, where optical throughput and detector responsivity drift but symbol timing does not.

The receiver streams per-pulse observations to a decoder that treats decoding as sequence estimation rather than classification. A hidden Markov model encodes the protocol’s byte framing directly in its state, so that transitions producing an inadmissible byte are pruned during search rather than filtered afterwards. Skip transitions represent undetected symbols, and inter-pulse gap timing supplies the evidence that makes them identifiable: a symbol that goes undetected lengthens the following gap by its own width plus one inter-symbol gap, indicating both *that* a symbol was lost and *which* one. No change to the transmitter, the receiver electronics, or the modulation is required: the improvement is obtained entirely from information the existing receiver already captures and discards.

The contributions of this work are:

- A characterization of a pulse-width OOK optical proximity link across five data rates from 28.6 to 171.4 bit/s, establishing that its dominant failure mode is undetected pulses rather than misclassified ones, and that every such omission nonetheless leaves a recoverable signature in the inter-pulse timing. This yields a reference level for what a decoder operating on the detected pulse stream can be expected to recover.
- A maximum-likelihood sequence decoder combining a learned emission model, gap-timing evidence, and the protocol’s framing and printable-ASCII constraints, which raises transmission success from 72.2% to 100% at 28.6 bit/s and from 36.1% to 69.4% at 114.3 bit/s on held-out data.
- A live closed-loop demonstration in which a memory-constrained receiver streams observations to the decoder and receives corrected messages back, raising transmission success from 5.9% to 44.9% at 114.3 bit/s over 118 transmissions, an architecture directly analogous to a surface asset offloading decoding to a better-resourced relay or lander.
- An account of the measurement pitfalls encountered on a resource-constrained receiver: an instrumentation artifact that inflated apparent symbol loss by as much as a factor of two and was mistaken for channel degradation until isolated, and a payload-length dependence that accounted for most of an apparent discrepancy between two experiments. A by-product of the first is a method for recovering the receiver’s detection-loop period, and with it a fixed edge-detection bias, from the periodicity of its own pulse-

width measurements.

The remainder of this paper is organized as follows. Section II reviews prior work in space optical communication, proximity links, receiver diversity, and sequence detection. Section III describes the link protocol, receiver hardware, and decoder model. Section IV details the experimental method and datasets. Section V presents the measurements, and Section VI concludes.

II. RELATED WORKS

Space optical communication research has concentrated overwhelmingly on *trunk* links: high-rate, long-range channels between spacecraft, or between a spacecraft and a ground station. This section reviews that literature and then the areas relevant to the opposite regime of short-range proximity links between co-located surface assets: short-range infrared signalling, multi-detector receiver front ends, maximum-likelihood sequence detection, and coding for channels that drop symbols. The last two supply the method adopted here; the first three establish the link model and the gap this work addresses. A closing subsection covers optical localization, the platform’s original objective, which is motivated but not evaluated in this paper.

A. Space Optical Communication

Prior work on free-space optical links establishes the performance benchmarks and design principles achieved by intensity-modulation direct-detection (IM/DD) systems across a wide range of distances. Hemmati [1] provides a systematic treatment of deep-space optical link architectures, deriving received power margins and pointing loss budgets as functions of aperture, beam divergence, and range. Biswas and Piazzolla [2] quantified these parameters for a Mars-to-Earth downlink, developing link-budget analyses based on received optical power and noise processes. At orbital distances, Sodnik et al. [3] demonstrated bidirectional optical communication in the European Space Agency SILEX program, validating acquisition, pointing, and tracking across an intersatellite link. Boroson et al. [4] reported the Lunar Laser Communication Demonstration, achieving a 622 Mbps downlink from lunar orbit using a laser transmitter and photon-counting receiver.

These systems share a common design posture. Range is large, the link budget is the binding constraint, and the receiver is a substantial instrument in its own right: cryogenically cooled detectors, precision gimbals, dedicated signal processing hardware. Performance is won through aperture, transmit power, and pointing accuracy.

Proximity links on a planetary surface invert every one of those assumptions. Range is metres, so link budget is comfortable. The receiver is a photodiode and a shared microcontroller on a mass- and power-constrained mobile asset. What limits the link is not how much optical power arrives but how much of the arriving signal a cheap receiver can successfully decode. The present work addresses that regime, which the trunk-link literature does not speak to.

B. Short-Range Infrared Links

Short-range optical wireless communication using light-emitting diodes has been characterized extensively for terrestrial indoor use, and that literature supplies the channel and receiver models applicable to surface proximity links. Gfeller and Bapst [5] established diffuse infrared as a practical in-building medium. Kahn and Barry [6] provided the foundational analysis of indoor infrared channels, deriving the tradeoffs among transmit power, receiver sensitivity, ambient-light shot noise, thermal noise, and achievable performance. Their treatment of IM/DD signalling, including pulse-position and pulse-width formats in which information is carried in the time domain rather than in amplitude, directly motivates the modulation used here. Ghassemlooy et al. [7] present a comprehensive treatment of channel modelling, path loss, multipath effects, and photodetector noise. Elgala et al. [8] survey the resulting system-level tradeoffs, including the tension among receiver field-of-view, angular sensitivity, and ambient-light rejection that governs photodiode front-end design.

Ambient-light rejection dominates much of this terrestrial work, and it is precisely the constraint that relaxes in a permanently shadowed lunar region, where solar infrared background is largely absent. What does *not* relax is the receiver's own limitation. Time-domain symbol encoding is the relevant inheritance: where an analog front end operates outside its linear region (as is measured to be the case for the platform described here), amplitude carries no usable information while symbol timing remains intact. The same argument applies to a surface deployment for different reasons, since regolith dust accumulation and thermal cycling degrade optical throughput and detector responsivity over a mission without disturbing symbol timing.

C. Multi-Detector Receivers

Receivers employing more than one photodetector have been studied primarily for multipath and ambient-light rejection. Carruthers and Kahn [9] analyse angle-diversity receivers in which multiple detectors with distinct orientations are combined, showing improved performance in non-directed links where a single wide field-of-view detector suffers co-channel interference and multipath dispersion. The receiver described here uses two photodiodes for a different purpose: detection robustness at the pulse level, with edge detection driven by the summed response so that a symbol arriving predominantly on one detector is not lost. On a mobile surface asset, where relative pointing between rover and lander changes continuously and cannot be assumed stable, this tolerance to partial illumination is a practical requirement rather than an optimisation. The normalized difference between channels, which cancels terms common to both, is noted as a bearing observable but is not evaluated here.

D. Maximum-Likelihood Sequence Detection

Recovering a transmitted sequence by searching for the most likely path through a trellis, rather than deciding each symbol independently, was formalized by Forney [10] for

channels with intersymbol interference. The underlying dynamic program is due to Viterbi [11], introduced for decoding convolutional codes and later given a general treatment by Forney [12]. The same recursion decodes hidden Markov models, whose canonical exposition is Rabiner's [13]: the observations are noisy and the underlying state sequence is unobserved.

A standard hidden Markov model pairs each state transition with exactly one observation, and it is that correspondence which the link measured here violates. Two established families relax it. Explicit-duration, or hidden semi-Markov, models allow a state to persist for a sampled duration rather than a single step [13]. Profile hidden Markov models, developed for biological sequence alignment, introduce explicit *delete* states that advance the chain while consuming no observation at all [14]. The decoder described in Section III is structurally of the latter kind: a pulse the receiver never detects still advances the byte framing, but produces nothing for the model to condition on.

That property is what makes the approach applicable here. The failure mode measured on this link is not symbol confusion but symbol *omission*: the receiver fails to detect a pulse, so no observation is produced for it. A per-symbol classifier has no mechanism to recover such a symbol, whereas a sequence model can represent it as an unobserved state transition and infer it from timing evidence and structural constraints.

E. Deletion Channels and Synchronization Errors

A channel that drops symbols without announcing it is, in coding-theoretic terms, a *deletion* channel, and codes for channels with insertion and deletion errors have been studied since Levenshtein [15]; Mercier et al. [16] survey the field. The closest prior work to the decoder used here is Davey and MacKay's watermark code [17], which embeds a known pseudorandom sequence in the transmitted stream and recovers synchronization with an inner hidden Markov decoder whose state space contains explicit insertion and deletion transitions. The mechanism this paper uses is therefore not new. A hidden Markov model with deletion transitions, decoded by dynamic programming, is the established tool for this channel, and the present decoder should be understood as an instance of it rather than as a new construction.

Two things differ, and both follow from the physical layer rather than from the algorithm. First, that literature generally assumes the transmitter can be designed: watermark and marker codes work by inserting redundancy for the receiver to synchronize against. Here the transmitter is fixed and nothing is added. The structure the decoder exploits, byte framing and a restricted payload alphabet, is pre-existing protocol overhead rather than deliberately inserted synchronization information, which is what makes the approach available as a retrofit to a deployed or mass-constrained system where the transmitter cannot be changed.

Second, and more consequentially, a deletion in the classical model is invisible. The receiver observes a shorter sequence

with no indication of where the loss occurred, and supplying that missing information is precisely why a watermark is inserted. In a pulse-width format the deletion is not invisible: it lengthens the following inter-pulse interval by a known, symbol-dependent amount, so the analog timing of the physical layer supplies for free the synchronization evidence a watermark exists to provide. A symbol-level deletion channel model discards exactly the quantity that makes recovery possible here.

The contribution of this work is accordingly empirical rather than theoretical. It reports a measurement, on hardware, that the timing side-information carried by a pulse-width physical layer is present in every single-omission transmission observed and is sufficient to localize and identify the lost symbol without added redundancy, together with a quantification of how much of the resulting message loss a sequence decoder actually recovers. Sequence detection is already well established in space communication in the form of convolutional coding with Viterbi decoding; what is measured here is what the same machinery buys when no redundancy is available to decode.

F. Optical Localization

Received optical signal strength has been investigated as a position observable where emitter positions are known *a priori*. Do and Yoo [18] survey visible-light positioning based on signal strength, angle of arrival, and time difference of arrival, and Lim [19] demonstrates three-dimensional LED-based positioning with sub-decimeter accuracy under line-of-sight conditions. Such techniques are attractive for surface operations because they would reuse an existing communication link as a navigation aid, avoiding dedicated sensing hardware in environments where GNSS is unavailable and visual odometry is degraded by low-contrast regolith and harsh shadowing. They are not evaluated in this paper: measurements reported in Section V show the receiver’s amplitude response to be flat with distance over the range tested, which precludes signal-strength ranging on the present front end and is addressed as future work.

III. SYSTEM OVERVIEW

LORAD comprises a fixed infrared beacon, a dual-photodiode receiver, and a decoder. This section describes the hardware, the link protocol, the receiver front end and detection logic, and the sequence model used for decoding.

A. Hardware

Both endpoints are Raspberry Pi Pico 2 W boards (RP2350) running MicroPython. They are physically separate and share no timebase or synchronization channel, so every timing relationship the decoder exploits is recovered from the optical signal alone.

The beacon drives a TSAL6200 infrared emitter through a 2N2222 common-emitter switch from GPIO 16. A second core runs an HTTP server that sets the broadcast message and the rate multiplier at runtime; the transmit loop reads both

once per message and is otherwise independent of network activity. Start pulses nominally 50 ms wide were *received* with a mean width of 50.18 ms and a standard deviation of 0.44 ms. That figure bounds transmit jitter rather than measuring it, since it also contains detection jitter and the receiver’s timing quantization, but at 0.9% of the symbol width it establishes that the concurrent server does not perturb transmit timing enough to matter.

The receiver carries two BPW34 photodiodes, each feeding an MCP6002 transimpedance stage with 100 k Ω feedback on a single 3.3 V rail, read on ADC0 and ADC1 (GPIO 26 and GPIO 27). The feedback resistor was corrected from 1 M Ω to 100 k Ω during development; all data reported here were collected after that change, with the two channels matched to a gain ratio of 1.15. The detectors are mounted close to coaxial rather than angled apart, which is why the normalized difference between them carries no usable bearing information at present (Section VI).

Two properties of the converter matter for interpreting the amplitude figures in Section V. The RP2350 samples with a 12-bit successive approximation converter, which MicroPython’s `read_u16` reports on a 16-bit scale; counts quoted out of 65,535 therefore carry 12 bits of real resolution. The converter is also multiplexed across both inputs, so charge from the previously selected channel biases each reading; the first conversion after each switch is discarded, at the cost of doubling the conversions per sample.

Detection runs as a MicroPython polling loop, and that loop’s period sets the receiver’s timing resolution. Fitting the periodicity of the 90,870 measured pulse widths gives a loop period of 262.4 μ s, or 3.81 kiloiterations per second, recovered independently at each of the five data rates and consistent across them to within 0.5%. This single number bounds the link. Symbol widths are quantized at roughly a quarter of a millisecond, and the shortest symbol spans 38 loop iterations at 1 $\times$ but only 6.4 at 6 $\times$ and 4.8 at 8 $\times$, which is the mechanism behind the symbol loss reported in Section V and the reason the useful rate range ends where it does.

B. Link Protocol

The beacon transmits baseband on-off keying. Information is carried in the *duration* of each emitted pulse rather than in its amplitude or in a modulated carrier. Three pulse widths form the symbol alphabet: 10 ms denotes a logical zero, 25 ms a logical one, and 50 ms a frame-start marker. Every pulse is followed by a fixed 10 ms idle period.

Fig. 1 shows one transmitted byte. Framing operates at byte granularity: each character is sent as a start pulse followed by its eight data bits, least-significant-bit first, giving nine pulses per byte. A message terminates with a newline, after which the transmitter idles for 500 ms and repeats, so a receiver may join the stream at any point and synchronize on the next start pulse. That rest is fifty times the inter-symbol gap, making message boundaries unambiguous even when individual symbols are lost.

One byte: symbol identity is pulse WIDTH, not amplitude

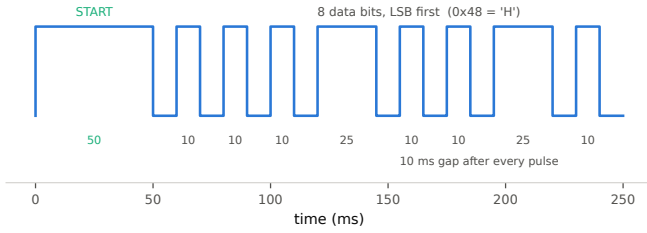


Fig. 1. One transmitted byte. Symbol identity is carried by pulse width, not amplitude. Each byte is preceded by a 50 ms start pulse and followed by eight data bits, least-significant-bit first; a 10 ms gap follows every pulse.

At nominal timing a byte occupies approximately 280 ms, of which 60 ms is framing overhead, giving a throughput of 28.6 bit/s. Two properties follow. First, the per-byte start pulse and the gap that follows it consume 21% of airtime. Second, because the zero and one symbols differ in duration, throughput is payload-dependent: an all-zero byte transmits in 220 ms against 340 ms for all-ones, a 55% spread. This dependence is revisited in Section V, where it accounts for a substantial part of the difference between two experiments.

To characterize the link across operating points, all four timing constants are divided by a common rate multiplier, preserving their ratios while compressing the symbol set in time. The multiplier is set at runtime and applied at message boundaries so that no transmission is split across two configurations. The firmware clamps the multiplier to the range $0.5\times$ to $8\times$, nominally 14.3 to 228.6 bit/s, but only $1\times$ through $6\times$ were characterized. The upper clamp is not a validated operating point: at $8\times$ a zero symbol spans 4.8 iterations of the $262.4\ \mu\text{s}$ detection loop, which is too few to time reliably.

C. Receiver

On the front end of Section III-A, rather than thresholding absolute signal level, which the DC bias of the photodiode network dominates, the receiver applies a first-difference high-pass filter with leaky integration,

$$y[n] = (x[n] - x[n-1]) + \alpha y[n-1], \quad \alpha = 0.85, \quad (1)$$

independently per channel, so that detection responds to signal transitions rather than to absolute intensity. A rising edge is declared when the summed filter output across both channels exceeds a threshold and a falling edge when it falls below its negation; the interval between them is the symbol duration.

Edges are declared on the *sum* of both channels rather than on either individually. Off-axis, the further detector may barely register, and a per-channel trigger would discard exactly those geometries.

The detection threshold adapts at runtime, tracking half the strongest recently observed edge under an exponential moving average, clamped to a fixed range and with a timeout that relaxes it by 10% per second if no edge is detected. No per-range tuning is required.

Serial output is buffered into preallocated arrays and emitted only during the inter-message rest. An earlier implementation wrote each pulse as it was detected, and because the sampling loop halts for the duration of a write, this cost measurable symbols at higher rates, an effect quantified in Section IV. Reads of the reverse channel are confined to the same idle window for the same reason.

D. Decoder

The decoder treats symbol recovery as sequence estimation. Hidden states are positions within the byte framing; observations are detected pulses. Fig. 2 shows both halves of the model: the framing chain, which is fixed by the protocol, and the three terms scored at each observation, of which two carry fitted parameters.

The model is a hidden Markov model and the recursion below is the ordinary Viterbi algorithm, but three departures from the textbook formulation $\lambda = (A, B, \pi_0)$ are worth naming, because each one simplifies or complicates the standard picture in a way that matters here.

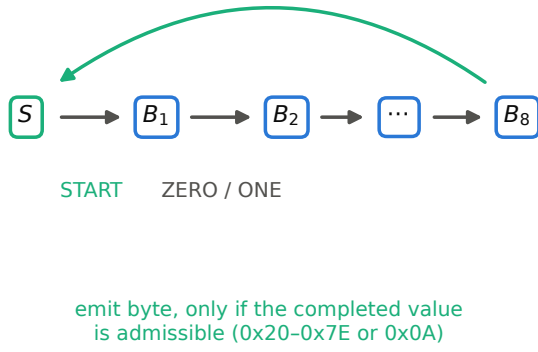
First, the transition structure A is not a learned stochastic matrix. It is a deterministic legality constraint read directly off the protocol: a transition either exists or it does not, and Fig. 2(a) is the whole of it. Nothing about the framing is estimated. Second, because the transmitted message is known exactly, parameter fitting is fully supervised and closed-form. No expectation-maximization over unlabeled sequences is required, so there are no local optima and no convergence criterion; the estimator is the sample mean and variance of a labeled set. Third, and most consequentially, a standard HMM emits exactly one observation per transition. Here a single observation may advance the chain by as many as $K + 1$ transitions, because symbols the receiver failed to detect still advance the framing even though they produced no pulse. The closest standard analogues are a profile HMM with delete states [14] and the inner decoder of a watermark code [17], and it is this departure that Fig. 2(b) is drawn to make visible: a skipped symbol is the one step in the trellis that descends two rows in a single column.

Viterbi rather than the forward recursion is used because the required output is a single self-consistent byte sequence, not a per-symbol posterior. A marginal confidence at each position would not be constrained to spell an admissible message.

1) *States and observations*: A state is the pair $q = (\text{bit position, partial byte value})$, giving $|Q| = 1 + \sum_{k=1}^8 2^{k-1} = 256$ states. Carrying the partial byte in the state allows the payload constraint to prune during search: a path that would complete an inadmissible byte is eliminated at the eighth data bit rather than producing a candidate that must be rejected afterwards. The admissible set is printable ASCII, 0x20 through 0x7E, together with the newline the beacon appends as its terminator, which must remain admissible for the final byte of a message to close.

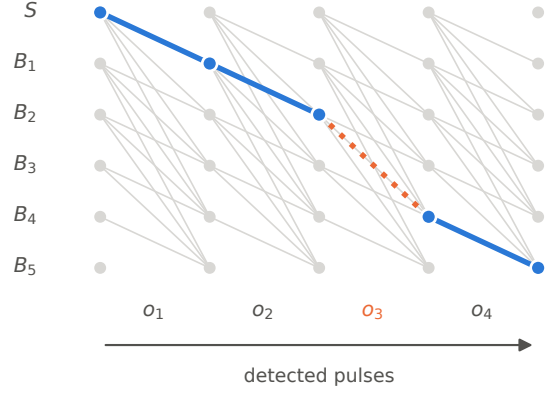
The t -th detected pulse contributes the observation $o_t = (d_t, g_t)$, where d_t is its measured width and g_t the interval since the previous falling edge. Write r for the rate multiplier,

(a) framing states and transitions, fixed by the protocol



each B_k is 2^{k-1} states carrying the partial byte;
256 states in total

(b) decoding as a trellis; bold is the surviving path



every edge scores $\log \mathcal{N}(d; \mu_s, \sigma_s^2) + \log \mathcal{N}(g; \hat{g}(k, u), ((1+k)\sigma_g)^2) + k \log \pi$

Fig. 2. The decoder. (a) The framing state machine, fixed by the protocol and not learned. (b) The same model unrolled into a trellis. Columns are detected pulses, rows are framing positions, grey edges are the legal advances, and the bold path is the one Viterbi retains. An ordinary pulse advances one framing position; the orange step at o_3 advances two on a single pulse, which is how a symbol the receiver never detected appears. Only μ_s , σ_s^2 , σ_g and $\log \pi$ are fitted from data.

$w(s)$ for the nominal width of symbol s at $1\times$, and G for the nominal inter-symbol gap.

2) *The three terms*: Each observation advances the chain by one emitting transition, optionally preceded by up to $K = 3$ skips $u = (u_1, \dots, u_k)$ representing symbols the receiver failed to detect. Three sources of evidence are combined in log space.

The *emission* term is a Gaussian class-conditional density over pulse width,

$$\log p(d_t | s) = -\frac{1}{2} \left[\log(2\pi\sigma_s^2) + \frac{(d_t - \mu_s)^2}{\sigma_s^2} \right], \quad (2)$$

with μ_s and σ_s^2 fitted per symbol and per rate. A generative form is used because the Viterbi recursion requires $p(o | s)$ directly, rather than a posterior that would have to be divided back through the class priors.

The *gap* term evaluates the observed interval against the interval the hypothesized skips predict. That prediction is structural, taken from nominal protocol timing rather than fitted,

$$\hat{g}(k, u) = (k+1) \frac{G}{r} + \sum_{j=1}^k \frac{w(u_j)}{r}, \quad (3)$$

and only its spread is learned:

$$\log p(g_t | k, u) = \log \mathcal{N}(g_t; \hat{g}(k, u), ((1+k)\sigma_g)^2). \quad (4)$$

The variance grows with k because a hypothesis that stacks several unobserved symbols accumulates their timing spread. Equation (3) is what makes a skipped symbol identifiable rather than merely permitted: its value depends on *which* symbols are hypothesized missing, not only how many.

The *skip prior* charges a fitted log-probability $\log \pi$ per unobserved symbol. Without a real cost here the decoder could posit arbitrarily many phantom symbols to force any byte into the admissible range.

3) *Recursion*: Let $\delta_t(q)$ be the best log-score of any path ending in state q having consumed t observations, with δ_0 zero at the start state and $-\infty$ elsewhere. Then

$$\delta_t(q') = \max_{q, u, s} \left[\delta_{t-1}(q) + k \log \pi + \gamma_t \log p(g_t | k, u) + \log p(d_t | s) \right], \quad (5)$$

maximized over source states q , skip sequences u of length $k \leq K$ that walk q to some intermediate state m along legal transitions, and symbols s whose transition carries m to q' . The gate $\gamma_t \in \{0, 1\}$ suppresses the gap term where it does not apply: at $t = 1$, because the first pulse of a transmission is preceded by the 500 ms inter-message rest rather than an inter-symbol gap, and wherever g_t exceeds $1.5 \hat{g}(K, \text{all START})$, the largest interval the skip model could legitimately explain. Decoding is the usual Viterbi backtrace from $\arg \max_q \delta_T(q)$, with beam pruning that discards paths trailing the leader by more than a fixed margin.

Fig. 3 illustrates the mechanism Equation (3) encodes: an undetected symbol lengthens the following gap by its own width plus one inter-symbol gap, so the observed gap indicates both that a symbol was lost and which one.

4) *What is fitted, and how*: Eight scalars are learned per rate: three emission means, three emission variances, one gap spread, and one skip prior. Everything else, the transition structure, the nominal widths $w(\cdot)$ and G entering (3), the

Transmitted

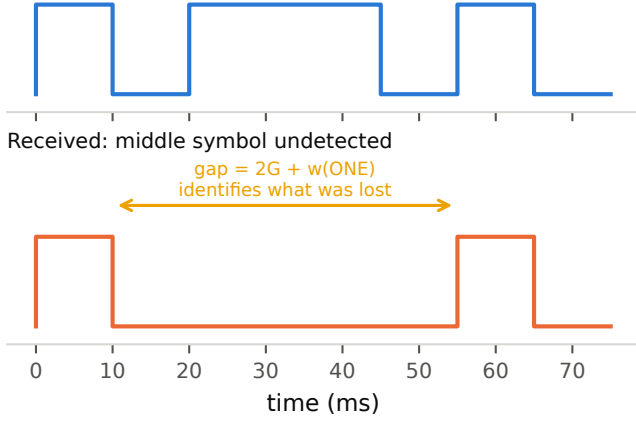


Fig. 3. Why an undetected symbol is recoverable. When the receiver misses a pulse, the following inter-pulse gap is lengthened by that pulse’s width plus one gap period, so its duration identifies the lost symbol.

skip bound K , and the beam width, is fixed by the protocol or chosen once and held. Table I lists the fitted values.

Emission parameters are maximum-likelihood estimates over training pulses labeled s ,

$$\begin{aligned}\mu_s &= \frac{1}{N_s} \sum_{i \in I_s} d_i, \\ \sigma_s^2 &= \max \left(\frac{1}{N_s} \sum_{i \in I_s} (d_i - \mu_s)^2, (0.02 \mu_s)^2 \right).\end{aligned}\quad (6)$$

The index set I_s is drawn only from training transmissions in which *every* symbol was detected. That restriction is what makes the labels exact: when the captured pulse count equals the transmitted symbol count, the i -th pulse is the i -th symbol by position, so no alignment heuristic is involved and the emission model cannot inherit an alignment error. The variance floor prevents a class measured as near-deterministic from dominating every path; it binds in 4 of the 15 cells, marked in Table I.

The gap spread is the sample standard deviation of the departure of observed gaps from nominal, over the same fully captured transmissions, floored the same way:

$$\sigma_g = \max(\text{sd}\{g_i - G/r\}, 0.02 G/r). \quad (7)$$

The skip prior is the observed capture shortfall, Laplace-smoothed:

$$\log \pi = \log \frac{M + \frac{1}{2}}{E + 1}, \quad (8)$$

where E is the total number of symbols transmitted across the training transmissions and M the number never detected. Smoothing is not cosmetic here. At 28.6 bit/s the receiver drops nothing at all, so the unsmoothed estimate is $\log 0$; the decoder must remain *able* to posit a skip, just very reluctant

TABLE I
FITTED PARAMETERS, PER RATE, FROM THE TRAINING MESSAGES ONLY. ALL QUANTITIES IN MS EXCEPT $\log \pi$. ENTRIES MARKED $\dagger$ ARE SET BY THE VARIANCE FLOOR OF (6) RATHER THAN BY THE DATA.

Rate (bit/s)	μ_s (σ_s)			σ_g	$\log \pi$
	ZERO	ONE	START		
28.6	10.18 (0.39)	25.17 (0.50) $\dagger$	50.17 (1.00) $\dagger$	0.38	−10.08
57.1	5.17 (0.35)	12.65 (0.27)	25.15 (0.50) $\dagger$	0.26	−5.80
85.7	3.50 (0.30)	8.47 (0.34)	16.82 (0.34) $\dagger$	0.29	−5.50
114.3	2.66 (0.21)	6.38 (0.32)	12.63 (0.40)	0.32	−5.54
171.4	1.84 (0.22)	4.31 (0.15)	8.47 (0.31)	0.25	−5.19

to. The smoothed value at that rate, $\log \pi = -10.08$, is almost five nats more expensive than at 171.4 bit/s, which is the model correctly learning that an unexplained symbol is far less plausible on a link that is not dropping any.

Table I also answers a fair objection to the emission model: if the protocol already specifies every symbol width, why learn μ_s at all? Because the receiver does not measure the transmitted width. Every fitted mean exceeds its nominal value, and by a strikingly stable amount: across all fifteen rate-symbol combinations the excess is $155 \pm 16 \mu\text{s}$, independent of both the symbol and the rate. That is 0.59 of the 262.4 μs detection loop period, exactly the signature of a falling edge being registered on the first loop iteration after it occurs. A decoder using nominal widths would place every class centre roughly half a sample low; fitting absorbs the bias without having to model its cause.

Because the decoder’s lattice requires a backpointer per state per observation (tens of thousands of entries for one message), it does not fit within the receiver’s memory, which raised allocation failures at 19 kB during development. The receiver therefore streams per-pulse observations over its existing USB serial connection and receives decoded messages back over the same link.

IV. EXPERIMENTS

Four experiments were conducted: a single-rate characterization of the front end and its error modes, a characterization of the link across data rates, an offline evaluation of the decoder against held-out transmissions, and a live closed-loop demonstration on hardware. All measurements were taken with the beacon and receiver in a fixed line-of-sight geometry at approximately 0.1 m, under ordinary indoor lighting. No result reported here varies the range, and none should be read as characterizing the link’s behaviour with distance.

A. Automated Data Collection

Labeled data were collected by driving the beacon over its network interface and capturing the receiver’s pulse stream. Because the transmitted message is set programmatically, the ground-truth symbol sequence is known exactly: no alignment heuristic is required, and there is consequently no risk of deriving labels from the same duration feature the model is later trained on.

The beacon re-reads its message at the start of each transmission, so a change takes effect only at a message boundary; the transmission in flight when the message changes is therefore discarded. Transmissions are separated in the captured stream by the 500 ms inter-message rest rather than by decoding them, keeping segmentation independent of the process under evaluation. Captures whose length departs substantially from the expected symbol count are rejected as partial or straddling a message change.

Payloads are drawn from three families. In the forty-message set used for the rate sweep, twenty-two messages are natural English prose sliced from a fixed corpus on word boundaries; ten are random printable ASCII drawn over letters, digits, punctuation and space; and eight are hand-constructed edge cases chosen to exercise framing, including runs of a single repeated character, alternating bit patterns, and bytes whose set bits sit at the ends of the least-significant-bit-first sequence. The three families are interleaved in transmission order so that payload type cannot become confounded with time. Retaining both prose and random payloads is deliberate: comparing them is what would separate a future language prior's contribution from what the structural constraints already provide.

B. Front-End and Error-Mode Characterization

A single-rate dataset was collected at 28.6 bit/s to characterize the front end and the baseline decoder's error modes with enough symbols to resolve rare events: 437 transmissions comprising 67,581 symbols, of which 283 carried natural text. Every symbol in this dataset was detected, which is what makes it usable for measuring misclassification in isolation from symbol loss. Per-pulse peak amplitude and peak filter response were recorded alongside duration.

C. Data-Rate Sweep

The link was characterized at five rate multipliers: $1\times$, $2\times$, $3\times$, $4\times$ and $6\times$, corresponding to 28.6, 57.1, 85.7, 114.3 and 171.4 bit/s. Forty distinct messages were transmitted at each rate, three captures retained per message, giving 597 labeled transmissions comprising approximately 91,000 pulses. The same forty messages were used at every rate so that differences are attributable to rate rather than payload. One capture was discarded by the analysis for carrying a serial line truncated mid-value.

Before each block the receiver's adaptive threshold is allowed to reconverge, and the serial buffer is flushed, so that captures made while the detector is still adapting do not enter the dataset.

Unless stated otherwise, every figure and table below is computed from this sweep. Where a quantity is available from more than one collection, the sweep is the one reported, so that no two numbers in a single comparison come from different runs.

D. Offline Decoder Evaluation

The decoder was evaluated against the threshold decoder implemented in the receiver firmware. Two aspects of the protocol make this comparison meaningful rather than rigged.

First, the baseline's acceptance windows are *scaled with the rate*. Leaving them at their nominal settings would cause the baseline to fail completely above $1\times$, which would be an artifact of the comparison rather than a result. The question posed is whether a fitted model outperforms hand-tuned thresholds that were themselves retuned for each operating point.

Second, train and test are split *by message*, never by capture. The three captures of a given message share a payload, so splitting at the capture level would place the same content on both sides of the split. Thirty percent of messages at each rate were held out, giving 36 test transmissions per rate and 35 at 171.4 bit/s. The eight parameters of Table I were fitted on the remaining 28 messages, independently at each rate.

The split is deterministic rather than random: messages are sorted and the first twelve held out. This has a consequence that must be stated, because it bears on how the offline numbers should be read. Sorting by character code places punctuation, digits and uppercase ahead of lowercase prose, so the held-out set contains *no* natural-text messages at all. All twelve are random or hand-constructed payloads, while 22 of the 28 training messages are English. Two things follow. The comparison between payload families motivated above cannot be performed on this split, and is left to future work. And the offline results are measured on payloads with no linguistic structure and a deliberate concentration of framing edge cases, which if anything understates the decoder relative to the traffic a deployed link would carry. Held-out messages also run longer than average, 174 symbols against 153 across the sweep, which matters in Section V.

1) *Metrics*: Two metrics are reported, and both are defined over *received transmissions* rather than over distinct payloads: one transmission is one complete emission of a message, so a message sent three times contributes three independent trials. Naming the unit this way is deliberate. The term *packet* would be misleading, because the link carries nothing of the kind in the networking sense, having no header, no length field and no checksum; and *message* alone would be ambiguous, because the same forty messages are transmitted repeatedly and at every rate. The unit of delivery is one newline-terminated message, and the unit of measurement is one received transmission of it.

Let transmission i carry the byte sequence $\mathbf{b}^{(i)} = (b_1, \dots, b_{L_i})$, comprising its payload characters and the newline the beacon appends as terminator, and let $\hat{\mathbf{b}}^{(i)}$ be what the decoder returns for it. Over N evaluated transmissions, the transmission success rate is

$$\text{TSR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{\mathbf{b}}^{(i)} = \mathbf{b}^{(i)}], \quad (9)$$

where $\mathbf{1}[\cdot]$ is the indicator function and equality is of the entire sequence. A single substituted, inserted or dropped byte fails the whole transmission. This is the quantity that matters to a link without retransmission, and it is deliberately unforgiving: no partial credit is given for a message that arrives nearly intact.

Character error rate measures how badly a failed transmission failed:

$$\text{CER} = \frac{\sum_{i=1}^N d_L(\hat{\mathbf{b}}^{(i)}, \mathbf{b}^{(i)})}{\sum_{i=1}^N L_i}, \quad (10)$$

where d_L is the Levenshtein distance. Three properties of (10) are worth stating because they affect how the numbers should be read. Levenshtein rather than Hamming distance is used because a decoder may return a sequence of a different length from the one transmitted, so insertions and deletions must be counted and not only substitutions. The ratio is a micro-average, formed from the aggregate edit distance over the aggregate transmitted length, so longer transmissions carry proportionally more weight than shorter ones. And a decode that fails outright, returning no admissible path, contributes $d_L = L_i$ and is charged as total loss rather than being excluded.

The term *character* error rate is used in preference to *byte* error rate to avoid collision with the conventional meaning of BER as bit error rate; since payloads are ASCII, one character is one byte throughout. The metric is identical in the offline and live evaluations.

A reference level for achievable performance is also computed. A transmission counts toward it if it was captured complete, or if it is missing exactly one symbol and the capture carries the signature that symbol would leave: an anomalous inter-pulse gap, or a merged pulse where a falling edge was missed. A transmission missing exactly one symbol that left no such signature is unrecoverable by any decoder operating on the detected pulse stream.

Two qualifications keep this from being a strict upper bound, and both make it conservative. The signature test is applied only to the exactly-one-missing class, so transmissions missing two or more symbols are counted as lost outright even though some may carry enough timing evidence to be recovered; on the held-out sets those number between 0 and 6 transmissions per rate. Insertions are likewise not credited. The level is therefore best read as what this decoder's mechanism can be expected to reach rather than as an information-theoretic limit, and a decoder exceeding it would not be a contradiction. It is evaluated on the held-out transmissions only, the same population the decoders are scored on, so that the three quantities plotted together in Section V describe the same set of messages.

E. Live Closed-Loop Operation

In the live configuration the receiver streams pulses continuously, a decoder process decodes each transmission as it completes and returns the decoded message over the same serial link, and the receiver maintains a log of received messages. Ground truth is polled from the beacon so that each decoded transmission is scored against the message current at the time it was decoded.

Live runs were conducted at 28.6 bit/s and 114.3 bit/s, yielding 39 and 118 transmissions respectively. The decoder

model was fitted from the sweep dataset at the corresponding rate and was not refitted during the run; the live traffic is therefore unseen data.

F. Instrumentation Validation

An artifact discovered during development is reported because it materially affected an earlier measurement and illustrates a hazard of instrumenting an embedded receiver.

In the original implementation the receiver wrote one line to its serial port per detected pulse. The sampling loop halts for the duration of a write, and the inter-symbol gap shrinks in proportion to the rate, so this dead time consumes an increasing share of the interval available to detect the next pulse. Taking a write to cost on the order of 1 ms, which is roughly four iterations of the measured 262.4 μ s detection loop but was not itself measured, the dead time would occupy 10% of the gap at 1 $\times$ and 60% at 6 $\times$. Measured symbol loss under that implementation rose over the same span from 0.21% to 0.88%, which is the signature of a fixed per-pulse dead time rather than a degrading optical channel.

Buffering the pulse records and emitting them during the inter-message rest reduced measured loss at every rate, though not uniformly: from 0.21% to zero at 1 $\times$, 0.37% to 0.32% at 2 $\times$, 0.59% to 0.44% at 3 $\times$, 0.76% to 0.37% at 4 $\times$, and 0.88% to 0.52% at 6 $\times$. As a ratio that is a factor of 1.15, 1.35, 2.04 and 1.71 at 2 $\times$ through 6 $\times$, with loss eliminated outright at 1 $\times$. The improvement is largest where the dead time occupies the largest share of the inter-symbol gap, but it reaches a full factor of two only at 4 $\times$. All results in Section V use the buffered implementation. The comparison is reported because the earlier measurements were initially interpreted as channel degradation, and the distinction was resolvable only by changing the instrumentation and repeating the experiment.

V. RESULTS AND EVALUATION

A. Amplitude Carries No Usable Information

The choice to encode symbols in the time domain is validated directly by the measurements. Received peak amplitude is pinned: across the 90,870 pulses of the sweep it sits at 41,370 of 65,535 converter counts with a standard deviation of 71 counts, a coefficient of variation of 0.17%, and the 67,581 pulses of the single-rate dataset give 42,171 counts at the same 0.16% spread. On a 3.3 V rail that peak corresponds to roughly 2.1 V, which is where an MCP6002 output saturates, so the invariance is the amplifier railing rather than a genuinely constant optical input.

The clearest evidence that the peak is uninformative is that it does not move even when the signal demonstrably does. Between 28.6 and 171.4 bit/s the peak high-pass response falls by 41%, from 1,742 to 1,033 counts, while over the same span the mean peak amplitude changes by 64 counts, or 0.16%. A quantity that holds to two parts in a thousand while the transition amplitude driving it changes by nearly half is carrying no information about the channel. An amplitude-discriminating scheme could not have functioned on this front end, and no signal-strength observable, for ranging or

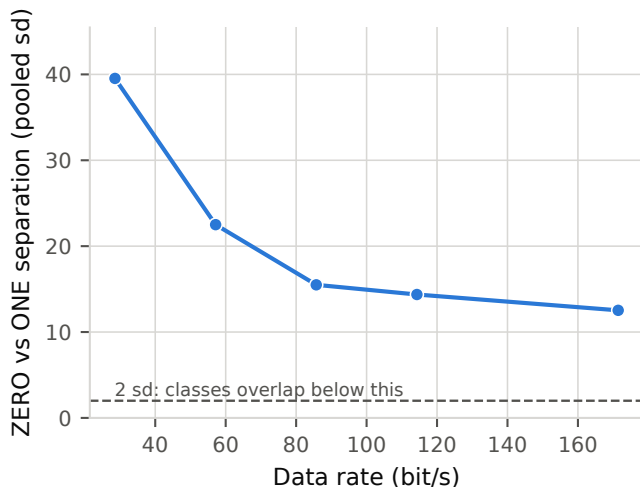


Fig. 4. Separation between the ZERO and ONE symbol classes as a function of data rate. Classes remain well separated at every rate tested, including rates at which decoding fails frequently.

for anything else, is available from it without reducing the transimpedance gain.

Pulse-width classes, by contrast, remain well separated. Fig. 4 shows the separation between the zero and one symbols in pooled standard deviations as a function of data rate. Separation falls from 39.5 at 28.6 bit/s to 12.5 at 171.4 bit/s, but even at the highest rate the classes are six times further apart than the two standard deviations at which distributions begin to overlap appreciably.

B. The Failure Mode Is Omission, Not Confusion

This separation has a direct consequence: symbol *misclassification* is not what limits the link. Over 67,581 symbols at 28.6 bit/s, no substitution errors occurred. The entire error budget consisted of 93 pulses (0.14%) whose durations fell between the threshold decoder's acceptance windows, and assigning each to its nearest class recovered all 93.

What limits the link is symbols the receiver never detects at all. At 114.3 bit/s, 0.371% of symbols went undetected, and 52.5% of transmissions were missing at least one. Because a byte spans nine pulses, even this sub-percent per-symbol rate compounds: it corrupts 3.3% of bytes, and over the 234 symbols of a 25-character message it leaves only 41.9% of transmissions intact. The compounding, not the per-symbol rate, is what makes the link unreliable, and it is the reason a metric averaged over symbols understates the problem by more than an order of magnitude.

Crucially, every undetected symbol left a recoverable signature. Of the transmissions missing exactly one symbol, 100% exhibited either an anomalous inter-pulse gap or a merged pulse consistent with a missed falling edge. This held at every rate tested, not only at 114.3 bit/s. The information required to reconstruct the lost symbol is present in the captured stream; a per-symbol classifier simply has no mechanism to use it.

TABLE II
OFFLINE EVALUATION, HELD OUT BY MESSAGE. APPROXIMATELY 36
TEST TRANSMISSIONS PER RATE.

Rate (bit/s)	Transmission success		Character error	
	Threshold	Sequence	Threshold	Sequence
28.6	72.2%	100.0%	1.44%	0.00%
57.1	11.1%	72.2%	6.32%	1.58%
85.7	16.7%	52.8%	5.75%	3.02%
114.3	36.1%	69.4%	3.88%	1.72%
171.4	31.4%	60.0%	4.03%	2.69%

C. Decoder Performance Across Data Rates

Table II and Fig. 5 report the offline evaluation on held-out messages. The sequence decoder outperforms the threshold decoder at every rate, by margins ranging from 28 to 61 percentage points of transmission success. It reduces character error rate at every rate as well, but by an amount that varies considerably: by half or better at 28.6, 57.1 and 114.3 bit/s, and by 48% and 33% at 85.7 and 171.4 bit/s respectively.

At 28.6 bit/s the decoder reaches 100% transmission success with zero character errors, matching the computed ceiling exactly: every error at that rate is a dead-zone boundary error, and all are eliminated.

Above that rate the decoder recovers between 42% and 57% of the headroom between the transmissions that arrived fully captured and the ceiling. At 57.1 bit/s, half the held-out transmissions arrived complete and the decoder raises transmission success to 72.2% against a ceiling of 88.9%; at 114.3 bit/s it reaches 69.4% from 47.2% complete, against a ceiling of 94.4%. These ceilings are computed on the held-out transmissions alone, so that all three quantities describe the same messages; evaluated over the full sweep instead, the same bounds read 93.3% and 95.8%, and the mismatch would flatter the decoder by up to nine points at 85.7 bit/s. The residual gap is genuine and is discussed in Section VI.

The dashed line in Fig. 5(a) is not monotonic in rate, and since that invites a search for a mechanism, it is worth stating exactly what sets it. Because every single-omission transmission carried a recoverable signature, the reference level reduces to an identity: it is one minus the fraction of transmissions that lost *two or more* symbols, and nothing else enters it. Across the five rates those transmissions number 0, 4, 6, 2 and 5 out of roughly 36.

Their ordering follows per-symbol loss exactly. Measured on the held-out transmissions alone, loss runs 0.000%, 0.367%, 0.495%, 0.335% and 0.448% (the 0.371% quoted earlier is the figure over the full sweep at 114.3 bit/s), and ranking the rates by loss reproduces the ceiling ordering inverted without exception: 114.3 bit/s loses fewer symbols than 85.7 bit/s, so its ceiling sits higher. Payload length cannot contribute, because the same twelve messages are held out at every rate. The ceiling is therefore a monotone function of per-symbol loss, and its non-monotonicity in *rate* is inherited from symbol loss being itself non-monotonic in rate.

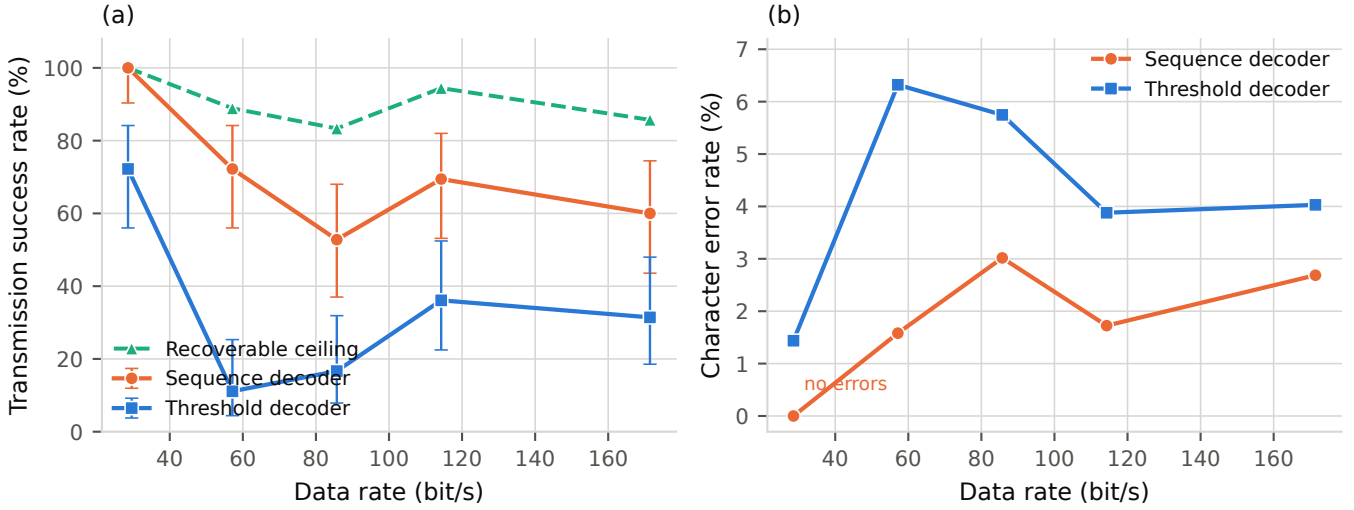


Fig. 5. Offline evaluation across data rates. (a) Transmission success rate, with 95% Wilson score intervals; the dashed line is the fraction of held-out transmissions that arrived complete or lost a single symbol that left a recoverable signature, a conservative reference level for a decoder operating on the detected pulse stream. (b) Character error rate. The sequence decoder made no errors at all at 28.6 bit/s.

The differences are nonetheless small counts. The step from 6 such transmissions at 85.7 bit/s to 2 at 114.3 bit/s carries Wilson intervals of 7.9–31.9% and 1.5–18.1%, which overlap heavily. The ceiling is best read as a level of roughly 85–95% above the lowest rate rather than as a curve whose shape is resolved.

The confidence intervals in Fig. 5(a) bear emphasis. With approximately 36 transmissions per rate, a point estimate near 70% carries an interval of roughly ± 15 percentage points. The two decoders are separated by non-overlapping intervals at 28.6, 57.1, 85.7 and 114.3 bit/s. At 171.4 bit/s they overlap, 18.6–48.0% against 43.6–74.4%, so the 28.6-point difference at that rate alone is suggestive rather than established. The finer structure of the per-rate curves is not resolved at this sample size and should not be read as a trend.

D. Live Closed-Loop Operation

Fig. 6 reports live operation with the decoder in the loop. At 28.6 bit/s over 39 transmissions the sequence decoder achieved 100% transmission success with zero character errors, against 69.2% and 1.54% for the threshold decoder. At 114.3 bit/s over 118 transmissions it achieved 44.9% against 5.9% (a factor of 7.6) and halved character error rate from 5.29% to 2.61%, recovering 46 of the 111 transmissions the threshold decoder lost. No pulses were reported dropped by the receiver in either run.

The live figures at 114.3 bit/s are lower than the corresponding offline figures, and the cause is payload length rather than any difference in the link or the decoder. The held-out messages that produced the offline result average 174 symbols; the live beacon transmitted a fixed 234-symbol message. At the measured per-symbol loss of 0.371%, the probability that a transmission arrives with no symbol missing falls from 52.4% to 41.9% between those lengths. Restricting the offline

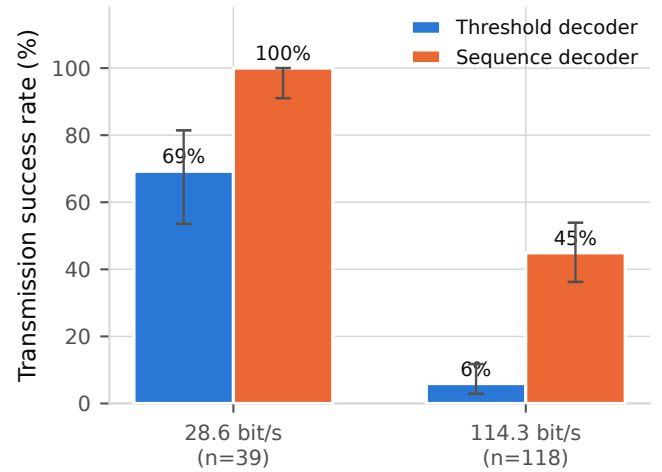


Fig. 6. Live closed-loop transmission success at two data rates, with 95% Wilson score intervals. The decoder ran on a host computer with the receiver streaming observations and receiving decoded messages over the same serial link.

data to messages of 200 symbols or more gives 19.4% of transmissions fully captured, so the live decoder's 44.9% is well above the fraction that arrived intact. This is the payload-length dependence noted in Section III appearing as a first-order effect, and it is the reason payload length must be held fixed or reported when comparing experiments.

E. Residual Error Characteristics

The decoder's remaining failures at 114.3 bit/s are almost exclusively single-character substitutions: *Brovn* for *Brown*, *Qudck* for *Quick*, *Quick\$Brown* for *Quick Brown*. When a symbol is undetected, gap timing localizes it and the printable-ASCII constraint narrows the candidates, but does

not always determine the answer uniquely: `Brown` is valid ASCII, so the structural constraint cannot reject it.

This error profile is informative. It indicates that the remaining headroom lies not in better timing evidence but in additional constraints on the payload itself, which motivates the language prior identified as the highest-value future extension.

VI. CONCLUSION

This paper presented LORAD, an experimental optical proximity link, and showed that replacing threshold decoding with maximum-likelihood sequence detection substantially improves reliability with no change to the transmitter, the receiver electronics, or the modulation. For a mass- and power-constrained surface asset, that distinction is the point: the improvement is obtained entirely from information the existing receiver already captures and discards, and costs only computation that need not reside on the asset itself.

The central measurement is that the link’s dominant failure mode is symbol *omission* rather than misclassification. Pulse-width classes remain separated by 12–40 pooled standard deviations at every rate tested, and at the lowest rate no substitution errors occurred across 67,581 symbols. What limits the link is symbols the receiver never detects: at 114.3 bit/s, 52.5% of transmissions were missing at least one, and because a byte spans nine pulses, a per-symbol loss of 0.371% compounds into majority message failure on a 25-character message. This has a direct architectural consequence: a better per-symbol classifier cannot help, because the missing observation was never produced. It is not visible from aggregate error rates alone.

Every undetected symbol nonetheless leaves a signature in the inter-pulse gap, present in 100% of the single-omission transmissions examined. A sequence decoder combining a learned emission model, gap-timing evidence, and the protocol’s framing and printable-ASCII constraints exploits that signature. It raises transmission success from 72.2% to 100% at 28.6 bit/s and from 36.1% to 69.4% at 114.3 bit/s on held-out data, and in live closed-loop operation from 5.9% to 44.9% at 114.3 bit/s, recovering 41% of the transmissions the threshold decoder lost.

Framed in mission terms, roughly two in five transmissions that would have required retransmission are instead recovered at first reception. On a proximity link operating within scheduled communication windows and a constrained energy budget, avoided retransmissions translate directly into window time and energy returned to the mission, and on a store-and-forward relay path they avoid delivery latency that can extend to an entire orbit.

Four limitations bound these results. First, the decoder recovers only 42% to 57% of the available headroom above the lowest rate (69.4% against a computed ceiling of 94.4% at 114.3 bit/s), and its residual failures are almost entirely single-character substitutions arising where the payload constraint admits more than one completion. Second, sample sizes of approximately 36 held-out transmissions per rate give confidence

intervals of roughly ± 15 percentage points; these separate the two decoders at four of the five rates but overlap at 171.4 bit/s, and they do not resolve the shape of the per-rate curves. Third, the held-out split contains no natural-language payloads, so the offline figures characterize the decoder on random and adversarially framed messages rather than on representative traffic, and the comparison between payload families that the dataset was built to support remains unmade. Fourth, and most consequentially for a flight application, the decoder does not fit in the receiver’s memory and runs on a host computer over the receiver’s serial link.

That third limitation is less restrictive than it first appears. The demonstrated architecture, a constrained endpoint streaming observations to a better-resourced decoder and receiving corrected messages back, mirrors a plausible surface topology, in which a rover offloads decoding to a lander or relay node with greater processing and thermal margin. Where that topology is unavailable, a fixed-lag decoder committing decisions a bounded number of symbols behind the observation front would bound memory and permit on-device operation. The remaining substitution errors are exactly what a character-level prior over the payload addresses, and the dataset deliberately mixes natural text with random ASCII so that such a prior’s contribution can be measured rather than assumed.

Two findings are methodological rather than architectural, and are reported because both were initially misread as properties of the channel. Instrumentation dead time (one serial write per detected pulse) inflated apparent symbol loss by up to a factor of two and produced a rate-dependent degradation closely resembling channel behaviour; only buffering the output and repeating the experiment separated the two. Separately, because the zero and one symbols differ in duration, throughput and failure probability both depend on the payload, and a difference in message length between two experiments accounted for most of an apparent discrepancy in their results. Both hazards are inherent to instrumenting a resource-constrained receiver, and both would be harder to diagnose on a deployed system than on a bench.

Finally, the platform’s original localization objective remains open and remains motivated: a surface asset that could derive bearing from a link it already maintains would gain a navigation aid without additional hardware, in an environment where GNSS is unavailable. Nothing here advances it, and one measurement stands in the way: the transimpedance stage saturates, pinning peak amplitude to within 0.17% across every pulse recorded, so the present front end yields no amplitude observable of any kind. The dual-photodiode receiver supports a normalized-difference bearing observable that would cancel range and gain, and the collection and analysis pipeline already accommodates angular labels; realizing it requires reducing the transimpedance gain so the amplifier stays linear, and angling the detectors apart, which are at present mounted close to coaxial. Whether amplitude varies usefully with range on a linear front end is untested: every measurement reported here was taken at a single fixed geometry.

ACKNOWLEDGMENT

The author acknowledges the use of Claude Code (Anthropic) as a development assistant throughout the design, implementation, and documentation of this project. Claude Code was used to support iterative development of embedded firmware, experimental design structuring, and signal processing implementation guidance. All final design decisions, experimental execution, and validation remain the responsibility of the author.

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